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North-West University

Engineering Design 2026 course team

[SYSTEM SPECIFICATION]

Hoisting System — Engineering Design 2026

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Origination and Approval

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Acceptance

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Approved by:			

Distribution List

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1 Introduction and scope

1.1 Identification

The system is a rail-mounted gantry/trolley hoisting platform, reusing the existing I-beam rail and gantry structure. A trolley moves laterally along the rail (MCTR I) and hoists a payload via a spooling mechanism (MCTR II), driven by an induction motor powered through custom drive electronics (EEII I). The payload is engaged via a custom electromagnetic attachment (EEII II). All subsystems interface with a shared embedded controller (ESP32-S3-WROOM-1 core board plus an isolated IO board), which executes real-time control locally and links wirelessly (nRF24Lo1) to a PC-based dashboard (REII) for monitoring, control, and motion-profile shaping.

The system decomposes into five work packages coordinated by this shared embedded controller, supplied to every group regardless of computer-engineering staffing: MCTR I (Lateral Movement), MCTR II (Spooling), EEII I (Drive Electronics), EEII II (Magnetic Load Attachment), and REII (Remote Control Dashboard).

1.2 Intended use

This document is an initial, high-level requirements specification for the 2026 Engineering Design hoisting system (Rev 0 — for discussion; bracketed [TBD] values are placeholders pending detailed design). It sits at the operational-to-technical level: it defines what the system and its work packages must do, not how each work package implements its solution — detailed component-level requirements are left to each work package's own design documentation (the Controller Specification, and the five work-package Scopes of Work listed in Section 2.1).

The system shall be operable via the PC-based dashboard where a group has computer-engineering capacity, and via the controller's standalone Wi-Fi/UDP interface where it does not (see 4.2.2 Modes).

1.3 Intended users

The system is intended to be built and operated by the Engineering Design 2026 student groups responsible for each work package, and demonstrated to course assessment staff at the end of the semester. Operators interact with the system either through the REII dashboard or, for groups without computer-engineering capacity, directly through the controller's standalone interface.

2 Applicable and other referenced documents

2.1 Applicable documents

Table 1: Applicable documents

Document Identifier	Document Description	Document Location
Controller Specification	Defines the shared embedded controller (ESP32-S3-WROOM-1 core board, IO board, isolation strategy, pin assignment) referenced throughout this specification.	Engineering Design 2026 project folder
MCTR I – Lateral Movement (SOW)	Work-package scope of work for the lateral movement mechanism.	Engineering Design 2026 project folder
MCTR II – Spooling (SOW)	Work-package scope of work for the hoist/spooling mechanism.	Engineering Design 2026 project folder
EEII I – Drive Electronics (SOW)	Work-package scope of work for the motor drive electronics.	Engineering Design 2026 project folder
EEII II – Magnetic Load Attachment (SOW)	Work-package scope of work for the electromagnetic load attachment.	Engineering Design 2026 project folder
REII – Remote Control Dashboard (SOW)	Work-package scope of work for the RF ground station and dashboard software.	Engineering Design 2026 project folder

2.2 Other referenced documents

Table 2: Other referenced documents

Document Identifier	Document Description	Document Location
Espressif ESP32-S3-WROOM-1 Datasheet	Governs GPIO count, strapping pins, and native-USB routing assumed by DR5 and the controller pin budget.	Espressif Systems (public)
Texas Instruments ADS1220 Datasheet	Governs the 2.048V internal reference used to bound single-ended current-sense signals in IR1/IR2.	Texas Instruments (public)
"2.2 EngDes26 – Requirements and Specs" (course slide deck)	Source of the requirements-engineering methodology (functional/performance/MOE/MOP/MOS taxonomy) applied throughout this document.	NWU Engineering Design 2026 course materials

3 Definitions, Acronyms, and Abbreviations

3.1 Definitions

As specified means as specified in Section 4 of this document.

May expresses permissive guidance.

Shall expresses a binding requirement.

State expresses an observable condition of the system at a given time.

Mode expresses behaviour exhibited by the system.

Work package means one of the five discipline-scoped deliverables (MCTR I, MCTR II, EEII I, EEII II, REII) into which the system's design and build effort is divided.

The Oxford Concise English Dictionary shall be used in the interpretation of terms in Section 4, Requirements, which are not otherwise defined above.

3.2 Acronyms

Acronym	Meaning
MCTR	Mechatronics (work-package prefix)
EEII	Electrical and Electronic Engineering II (work-package prefix)
REII	Remote-control Engineering / dashboard work package
FR	Functional Requirement
NFR	Non-Functional Requirement (superseded in this document by Other Qualities / Physical requirements per template structure)
DR	Design Requirement
IR	Interface Requirement
PT	Performance Target
MOP	Measure of Performance
MOS	Measure of Suitability
ADC	Analog-to-Digital Converter
PWM	Pulse-Width Modulation
RF	Radio Frequency
RCD	Residual-Current Device
COTS	Commercial Off-The-Shelf
UDP	User Datagram Protocol
HV	High Voltage

Acronym	Meaning
AC / DC	Alternating Current / Direct Current
NWU	North-West University
TBD	To Be Determined

3.3 Abbreviations

None beyond the acronyms listed in 3.2.

4 Requirements

4.1 Identification of external interfaces

Interface	Description
Controller ↔ EEI I (Drive Electronics)	Isolated 6-line gate/PWM interface plus one reserve line; single-ended current-sense input. See IR1.
Controller ↔ EEI II (Magnetic Load Attachment)	Isolated PWM interface and single-ended current-sense input ($\leq 2.048V$). See IR2.
Controller ↔ MCTR I (Lateral Movement)	Direction, PWM, and quadrature-encoder interface. See IR3.
Controller ↔ MCTR II (Spooling)	Load-cell input and brake-control interface. See IR4.
Controller ↔ REI (Dashboard)	Wireless nRF24Lo1 link carrying telemetry and control commands; comms protocol is REI's own design.
Controller ↔ Expansion header	5-pin, protocol-agnostic, low-voltage digital logic header for group-specific extensions. See IR5.

4.2 Identification of states and modes

4.2.1 States

- Idle — system powered but not executing a commanded motion, hoist, or magnet-engagement action.
- Operational — system actively executing a commanded lateral, hoist, or magnet-engagement function under either mode below.
- Fault / E-Stopped — the hardware e-stop has removed power from the HV bus (see PHY2); all motive functions are inhibited independent of software or firmware state.

4.2.2 Modes

- Dashboard-controlled — the operator controls the system via the REI PC-based dashboard over the nRF24Lo1 link.
- Standalone — for groups without computer-engineering capacity, the operator controls the system directly via the controller's built-in Wi-Fi/UDP interface, without a dashboard.

4.3 System function and performance requirements

Requirements are grouped by owning work package, and split across tables so each stays readable on one page. Each requirement is a single, testable statement per the parsing-template discipline (actor, condition, action, object, constraint). Where a requirement has a quantitative performance dimension, it is deliberately written as a functional pass/fail threshold only — the associated performance target is separated out below. This reflects experience that functional correctness, not raw performance, is the harder and more common failure point: a system that merely functions should pass, with performance differentiating marks rather than gating them.

4.3.1 Mechanical (MCTR I, MCTR II)

ID	Requirement
FR1	The lateral movement mechanism, upon receipt of a motion command from the controller, shall move the trolley along the I-beam rail in the commanded direction, using a COTS DC motor costing less than R250 (MCTR I). See PT1 for the associated performance target.
FR2	The lateral movement mechanism shall report wheel position to the controller via quadrature encoder feedback, to support motion profiling by REII (MCTR I).
FR3	The lateral movement wheels shall be custom-manufactured by 3D printing, supporting the wheel wear-characterization exercise (MCTR I).
FR4	The spooling mechanism, upon receipt of a raise or lower command, shall drive the hoist gearbox to lift or lower a payload of up to 20kg in the commanded direction (MCTR II). See PT2 for the associated performance target.
FR5	The spooling mechanism shall report on-hook load to the controller continuously during operation (MCTR II).
FR6	The active brake shall engage automatically upon loss of drive power or command signal, holding the payload without drift (MCTR II).

4.3.2 Electrical (EEII I, EEII II)

ID	Requirement
FR7	The drive electronics, upon receipt of gate/PWM commands from the controller, shall convert single-phase 230VAC mains to a DC bus and invert this to a 3-phase drive for the induction motor (EEII I).
FR8	The drive electronics shall detect and interrupt drive output upon shoot-through or overcurrent conditions, protecting the switching devices from damage (EEII I).
FR9	The drive electronics shall provide a single-ended current-sense signal, capped below 2.048V, to the controller for firmware-based fault inference (EEII I).
FR10	The magnetic load attachment, upon receipt of an engage command, shall generate sufficient holding force via a 3-point electromagnet to retain a payload of up to 20kg (per DR6) under static conditions (EEII II). See PT3 for the associated performance target.
FR11	The magnetic load attachment shall report an on-load indication to the controller, derived from current-sense discrimination between a loaded and unloaded grip (EEII II).

4.3.3 Software and Controller (REII, cross-cutting)

ID	Requirement
FR12	The dashboard, upon receipt of telemetry from the controller via the nRF24Lo1 link, shall display all operational parameters (position, speed, load, current, fault, grip status) to the operator (REII).
FR13	The dashboard shall provide the operator full control of all system operations, including motion, hoist, and magnet-engagement commands (REII).
FR14	The embedded controller shall execute its real-time control loop — drive signal generation, sensor acquisition, safety interlocks — independently of wireless link or dashboard availability.
FR15	For groups without computer-engineering capacity, the controller shall accept motion and control commands via UDP over its built-in Wi-Fi, requiring no additional hardware.

4.3.4 Performance targets

Graduated marking criteria, not pass/fail gates — each links back to a functional requirement above. Marks scale with how closely the target is achieved rather than requiring it outright. Consistent with treating these as Measures of Performance (directly quantifiable, expressible on a scale) rather than binary compliance checks.

ID	Linked FR	Target	Marking Basis
PT1	FR1 (Lateral movement)	≥ 1 m/s, measured at 1m from the departure point, unloaded	Full marks at target; marks scale down proportionally for lower demonstrated speed.
PT2	FR4 (Spooling)	1.6m hoist stroke in ≤ 20 s (≈ 80 mm/s), with a 20kg payload	Full marks at target; marks scale down proportionally for a slower demonstrated rate.
PT3	FR10 (Magnetic attachment)	Retain a 20kg payload against a disturbance force [TBD]	Full marks at target; marks scale down proportionally below threshold. Numeric threshold pending further dynamics modelling.

4.4 Relationship between states and modes

Both operating modes (Dashboard-controlled and Standalone) are available whenever the system is in the Operational state; mode selection does not affect what states the system can enter. The Fault/E-Stopped state pre-empts both modes, inhibiting motive output regardless of which mode was active when the e-stop engaged. The system begins in the Idle state at power-up, before an operator issues a first command under either mode.

4.5 System external interface requirements

Listed in the same order as the interfaces identified in 4.1.

ID	Requirement
IR1	The controller shall present an isolated 6-line gate/PWM interface to the drive electronics for 3-phase motor control, plus one reserve line.
IR2	The controller shall present an isolated PWM interface and a single-ended current-sense input ($\leq 2.048\text{V}$) to the magnetic load attachment.
IR3	The controller shall present a direction, PWM, and quadrature-encoder interface to the lateral movement mechanism.
IR4	The controller shall present a load-cell input and a brake-control interface to the spooling mechanism (channel to be finalized — ADS1220 differential input or dedicated HX711).
IR5	The controller shall present a 5-pin, protocol-agnostic expansion header for group-specific extensions, documented as low-voltage digital logic only.

Full pin-level detail (module pins, edge-header assignment, isolation strategy) is maintained in the Controller Specification document, not duplicated here.

4.6 Environmental requirements

Deployment: the system shall operate on the existing I-beam rail and gantry structure, located outdoors on the North-West University North Campus.

The system shall operate outdoors, exposed to ambient weather conditions typical of the NWU North Campus — temperature variation, precipitation, solar/UV exposure, wind, and dust — and in the presence of other groups' concurrently operating systems on shared or adjacent rail infrastructure.

Electrical enclosures and wiring shall provide ingress protection against rain, dust, and incidental moisture, given the system's outdoor deployment; exposed materials, including any 3D-printed or polymer components, shall be selected or rated for sustained outdoor UV and temperature exposure across the semester.

Operational life-cycle: the system shall remain operational through the design, build, integration, and demonstration phases of the 2026 academic semester.

4.7 External resource requirements

Maintains the same order as 4.1 / 4.5 — resources enter or exit the system only via the interfaces already identified.

The system shall receive electrical power via a single dedicated 1-phase, 50Hz, 230VAC mains supply point per group. Distribution of this supply to all subsystem voltage rails (e.g. 3V/5V/12V/24V, each independently selected COTS supplies by their owning work package) and provision of the main on/off switch is a system-level integration responsibility, assigned to EEII I.

4.8 Physical requirements

ID	Requirement
PHY1	The payload used to test and demonstrate the system shall present a flat sheet-metal attachment surface of 600mm x 400mm, at least 3mm thick, for engagement by the magnetic load attachment. This surface may be supplemented with additional mass to raise the total payload to a maximum of 20kg, provided the combined centre of mass remains centred on the 600mm x 400mm attachment surface. (Formerly DR6; the general system mounting plate is unaffected by this requirement and is not required to be sheet metal.)
PHY2 (Safety)	The system shall include a hardware e-stop wired directly into the HV bus, capable of removing power from the drive electronics independent of software or firmware state.
PHY3 (Safety)	The mains supply feeding the drive electronics shall be protected by a residual-current device (RCD), given the combination of HV wiring and outdoor exposure to moisture.

4.9 Other qualities

ID	Requirement
OO1 (Reliability)	Shoot-through and overcurrent protection shall be validated on a milled test board prior to full-scale VFD assembly, directly addressing the prior year's dominant failure mode.
OO2 (Maintainability)	The core controller board shall be designed as reusable general-purpose lab infrastructure, providing local analog acquisition and data logging independent of any single year's design problem.
OO3 (Usability)	The dashboard interface shall present telemetry and controls such that an operator can safely command the system after a brief orientation, with no specialist RF or embedded-systems knowledge required.
OO4 (Affordability)	Component costs shall not exceed the stated constraints for each work package (e.g. lateral-movement motor < R250); total system cost per group shall not exceed R5,000, inclusive of all work-package procurement.

4.10 Design and construction requirements

Design requirements constrain the solution space ahead of detailed design. Per requirements-engineering practice, each is included only with an explicit rationale, since design requirements should be used conservatively.

ID	Requirement	Rationale
DR1	The lateral movement drive motor shall be a COTS DC motor costing less than R250.	Bounds procurement cost and lead time across all groups; keeps motor selection a sizing exercise rather than an open-ended search.
DR2	The spooling gearbox shall not use a worm gear configuration.	Forces a multi-stage gearbox design exercise consistent with the programme's mechanical design intent, rather than a single high-ratio COTS-equivalent solution.
DR3	The magnetic load attachment shall not use a complete COTS electromagnet or solenoid; COTS core material is permitted.	Preserves hands-on electromagnetic design and manufacturing experience while keeping raw-material procurement low-risk.
DR4	The wireless link shall use the nRF24Lo1 module.	Standardizes the wireless link across every group and the provided controller firmware, so the fixed low-level link behaviour is consistent everywhere.
DR5	The embedded controller shall be based on the ESP32-S3-WROOM-1 (N8R2) module.	Standardizes the platform provided to every group regardless of computer-engineering staffing, ensuring consistent GPIO and ADC capability.

5 Notes

5.1 Verification requirements

Each requirement above will be assigned an explicit verification method (how compliance is measured, and with what) in the next revision of this document, following the Measure of Performance / Measure of Suitability framework: the Section 4.3.4 performance targets (PT1–PT3) are Measures of Performance; supportability-type requirements (e.g. OQ1's protection validation, OQ2's maintainability) are Measures of Suitability. Functional requirements in Section 4.3 are verified as pass/fail — demonstrated or not — independent of the graduated scoring applied to their linked performance targets. Verification requirements will be written with the same rigor as the requirements themselves — explicit, measurable, and tied to a defined test method or standard — rather than left implicit.

5.2 List of safety requirements

Duplicated from Section 4.8 (Physical requirements) for visibility; see 4.8 for the authoritative statement.

- PHY2 (Safety) — The system shall include a hardware e-stop wired directly into the HV bus, capable of removing power from the drive electronics independent of software or firmware state. (See 4.8.)
- PHY3 (Safety) — The mains supply feeding the drive electronics shall be protected by a residual-current device (RCD), given the combination of HV wiring and outdoor exposure to moisture. (See 4.8.)

5.3 Open items

The disturbance-force threshold in PT3 remains open pending further dynamics modelling. Payload mass is fixed at up to 20kg per PHY1, and the lateral/hoist performance targets are fixed per PT1/PT2. The spooling load-cell's ADC path (ADS1220 differential channel vs. dedicated HX711) remains an open implementation choice per the Controller Specification and is not treated as a hard requirement here.

Document History

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0	2026-08-04	For discussion	System_Requirements_Specification_2026

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0	2026-08-04	Initial high-level draft, per the Engineering Design 2026 requirements-engineering methodology.

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